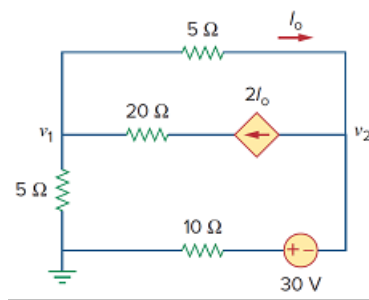


## Problem

For the circuit in Fig. 3.70, find  $v_1$  and  $v_2$  using nodal analysis.

**Figure 3.70:**

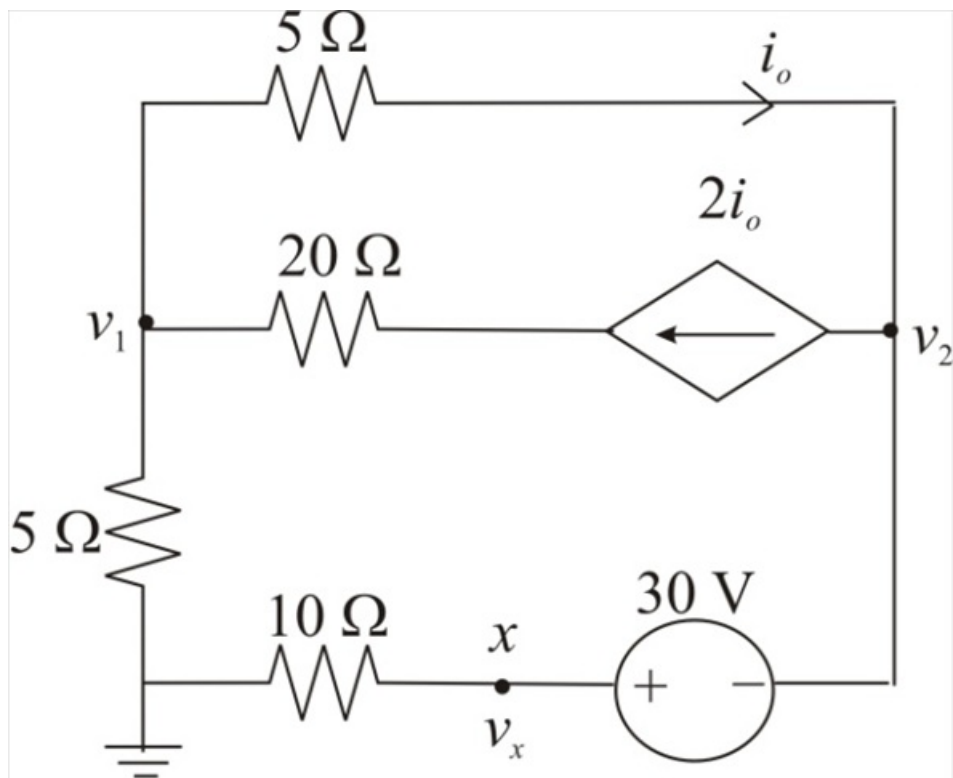


## Step-by-step solution

## Step 1 of 5

Refer to Figure 3.70 in the text book.

Identify the nodes and voltages in the circuit shown in Figure 1.



**Figure 1**

## Step 2 of 5

Notice that the nodes 2 and x represent super node.

Consider the expression for the current  $i_o$ .

$$i_o = \frac{v_1 - v_2}{5}$$

Therefore, the expression for the current  $i_o$  is,  $\frac{v_1 - v_2}{5}$ .

### Step 3 of 5

Consider the relation between the voltage  $v_x$  and  $v_2$ .

$$v_x - v_2 = 30$$

$$v_x = 30 + v_2$$

Therefore, the expression for the voltage  $v_x$  is,  $30 + v_2$ .

### Step 4 of 5

Apply nodal analysis at node 1.

$$\frac{v_1}{5} + i_o - 2i_o = 0$$

$$\frac{v_1}{5} - i_o = 0$$

Substitute  $\frac{v_1 - v_2}{5}$  for  $i_o$ .

$$\frac{v_1}{5} - \left( \frac{v_1 - v_2}{5} \right) = 0$$

$$\frac{v_1}{5} - \frac{v_1}{5} + \frac{v_2}{5} = 0$$

$$\frac{v_2}{5} = 0$$

$$v_2 = 0 \text{ V}$$

Therefore, the value of the voltage  $v_2$  is, 0 V.

### Step 5 of 5

Apply super node analysis at node 2 and x.

$$\frac{v_x}{10} - i_o + 2i_o = 0$$

$$\frac{v_x}{10} + i_o = 0$$

Substitute  $30 + v_2$  for  $v_x$  and  $\frac{v_1 - v_2}{5}$  for  $i_o$ .

$$\frac{30 + v_2}{10} + \frac{v_1 - v_2}{5} = 0$$

Substitute 0 V for  $v_2$ .

$$\frac{30 + 0}{10} + \frac{v_1 - 0}{5} = 0$$

$$\frac{30}{10} + \frac{v_1}{5} = 0$$

$$3 + \frac{v_1}{5} = 0$$

$$\frac{v_1}{5} = -3$$

$$v_1 = -15 \text{ V}$$

Therefore, the value of the voltage  $v_1$  is, -15 V.